

Problem Session III

Warm-up question

What is the difference between conditional and unconditional probability?

Question 1

A university surveyed students about whether they own a car and whether they live on campus.

The following information was collected:

60% of students own a car. 15% of students both own a car and live on campus. 43% of students live on campus.

Construct a probability tree diagram (see page 6 of your guided notes) based on this information.

Question 2

A department store sells sport shirts in three sizes (small, medium, and large), three patterns (plaid, print, and stripe), and two sleeve lengths (long and short). The accompanying table gives the proportions of shirts sold in the various category combinations.

Length	Size	Plaid	Print	Striped
Short	Small	0.04	0.02	0.05
Short	Medium	0.08	0.07	0.12
Short	Large	0.03	0.07	0.08
Long	Small	0.03	0.02	0.03
Long	Medium	0.10	0.05	0.07
Long	Large	0.04	0.02	0.08

- (a) What is the probability that the next shirt sold is a medium, long-sleeved, print shirt? (2 points)
- (b) What is the probability that the next shirt sold is a medium print shirt? (4 points)
- (c) What is the probability that the next shirt sold is a long sleeved shirt? (4 points)
- (d) What is the probability that the next shirt sold has a print pattern? (4 points)

- (e) If the next shirt sold was a short sleeved plaid, what is the probability it was of medium size? (4 points)
- (f) If the next shirt sold was a medium plaid, what is the probability it was long sleeved? (4 points)

Question 3

What would make this an invalid joint probability distribution?

Try and think of two potential ideas...

Length	Size	Plaid	Print	Striped
Short	Small	0.04	0.02	0.05
Short	Medium	0.08	0.07	0.12
Short	Large	0.03	0.07	0.08
Long	Small	0.03	0.02	0.03
Long	Medium	0.10	0.05	0.07
Long	Large	0.04	0.02	0.08

Question 4

A city survey categorizes households by pet ownership and activity level:

Group	Highly Active	Not Highly Active
Dog owners	0.30	0.20
Cat owners	0.15	0.15
No pets	0.05	0.15

- Compute the probability that a randomly selected household is highly active.
- Compute the probability that a household owns a dog given that it is highly active.